

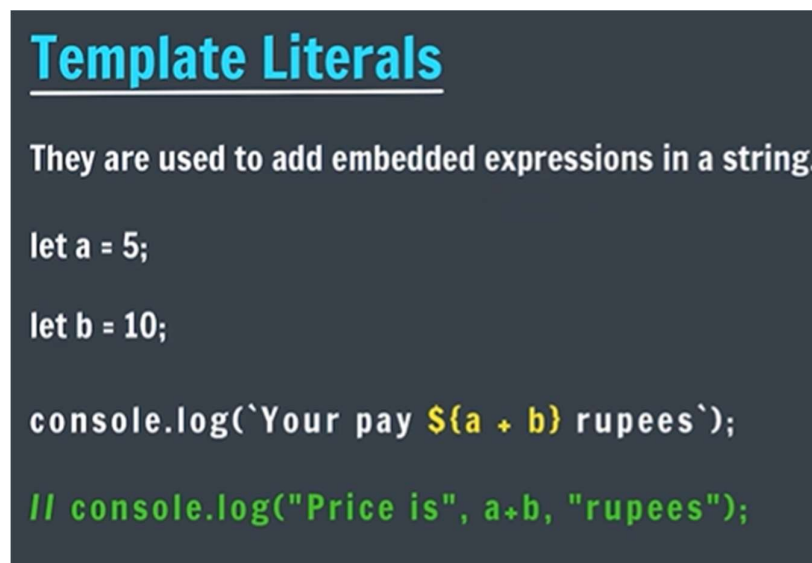
→JS-2

Linking JS file

→by using the script tag in the head of the html we add the link of js file

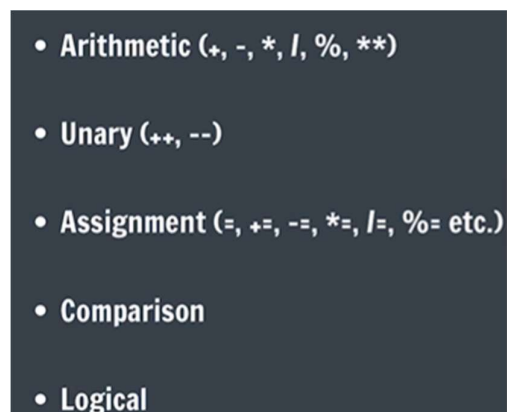


→Template Literals

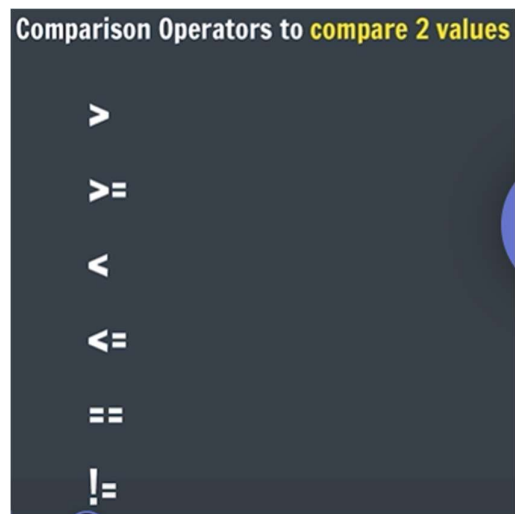


→`\${}`→what ever written inside it will calculate and then it will give the final output it will print in console.

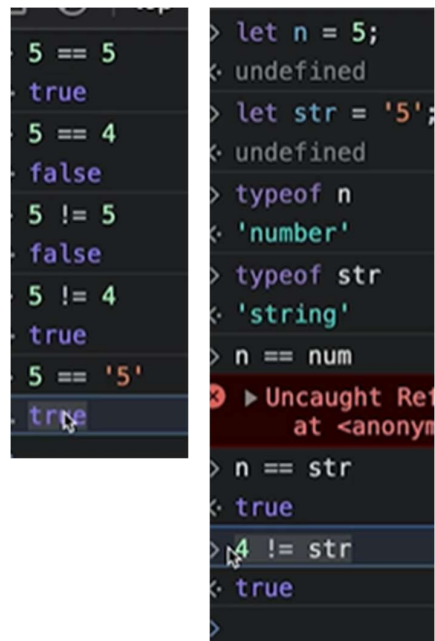
→Operators in JS



→ Comparison Operators



== → it compares the values in it not the data type



→ === → it compares the both value and type

==

- compares value, not type

```
> "123" == 123
< true
> 1 == '1'
< true
> 0 == ''
< true
> 0 == false
< true
> null == undefined
< true
```

===

- compares type & value

```
> "123" === 123
< false
> 1 === '1'
< false
> 0 === ''
< false
> 0 === false
< false
> null === undefined
< false
```

→ Comparison for non-numbers

```
> 'a' > 'A'
< true
> 'a' > 'b'
< false
> 'b' < 'c'
< true
> 'B' < 'C'
< true
> '*' < '&'
< false
```

'a' -> 61, 'b' -> 62 ...
'A' -> 41 'B' -> 42 ...

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→ Conditional Statements

If else

Nested if-else

Switch

→if statement

if Statement

// some code before if

if (some condition) {

// DO SOMETHING

}

// some code after if

```
//Conditional Statements
console.log("before my if statement");
let age = 14;
if (age >= 18) {
  console.log("you can vote");
  console.log("you can drive");
  let a = 5;
  console.log(5 * a);
}
console.log("after my if statement");
```

```
//Conditional Statements
console.log("before my if statement");
let age = 23;
if (age >= 18) {
  console.log("you can vote");
  console.log("you can drive");
  let a = 5;
  console.log(5 * a);
}
console.log("after my if statement");
```

→ whatever you write inside the if loop it will executed only when the if condition is true

→else if statement

else if Statement

if (condition1) {

// DO SOMETHING

}

else if (condition2) {

// DO SOMETHING ELSE

}

else if (condition3) {

// DO SOMETHING ELSE

}

→ Else

else Statement

```
if (condition1) {
```

```
    // DO SOMETHING
```

```
}
```

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```
else {
```

```
    // DO SOMETHING ELSE
```

```
}
```

→ Nested if-else

Nested if-else

Nesting is writing if-else inside if-else statements. It can have many levels.

```
if marks >= 33
```

```
    if marks >= 80
```

```
        print "O"
```

```
    else
```

```
        print "A"
```

```
else
```

```
    print "better luck next time!"
```

→ Logical Operators

Logical Operators

Logical Operators to **combine expressions**

&& Logical AND

(exp1) && (exp2)

```
> true && true
< true
> true && false
< false
> false && true
< false
> false && false
< false
```

```
(5 > 3) && (3 > 1)
true
(5 > 3) && (3 < 1)
false
(5 < 3) && (3 > 1)
false
(5 < 3) && (3 < 1)
false
```

→ Logical OR

→ NOT operator

```
top ▾
> true || true
< true
> true || false
< true
> false || true
< true
> false || false
< false
>
```

```
!true
false
!false
true
```

→truthy & falsy

truthy & falsy

Everything in JS is true or false (in boolean context).

This doesn't mean their value itself is false or true, but they are treated as false or true if taken in boolean context.

Falsy values

false, 0, -0, 0n (BigInt value), "" (empty string), null, undefined, NaN

Truthy values

Everything else

```
if (true) {  
  console.log("it has true value");  
} else {  
  console.log("it has false value");  
}  
  
if (" ") {  
  console.log("it has true value");  
} else {  
  console.log("it has false value");  
}
```

top
it has false value
it has true value

```
pp.js > ...  
let string = " ";  
  
if (string) {  
  console.log("string is not empty");  
} else {  
  console.log("string is empty");  
}  
  
let num = 0;  
  
if (num) {  
  console.log("num is not equal to 0");  
} else {  
  console.log("num is equal to 0");  
}
```

string is not empty
num is not equal to 0

→ Switch Statement

Switch Statement

Used when we have some fixed values that we need to compare to.

```
let color = "red";

switch(color) {
  case "red" :
    console.log("stop");
    break;
  case "yellow" :
    console.log("slow down");
    break;
  case "green" :
    console.log("GO");
    break;
  default :
    console.log("Broken Light");
}
```

→ Alert and Prompt

Alert & Prompt

Alert displays an alert message on the page.

```
alert("something is wrong!");
```

Prompt displays a dialog box that asks user for some input.

```
prompt("please enter your roll no.");
```

```
5 app.js > [⌘] msg
1   let firstName = prompt("enter first name");
2   let lastName = prompt("enter last name");
3   let msg = "Welcome " + firstName + " " + lastName + "!";
4   alert(msg);
5
```